Title
Gene Expression Inference from Adalimumab Immunogenicity Outcomes in Rheumatoid Arthritis

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Abstract

Autoimmune disease happens when cells in the body mistakenly attack healthy cells in an immune response. Many therapies have been developed to treat a variety of autoimmune diseases; one particular family of drugs, biologics, involves engineered proteins that target inflammation to treat these diseases. Adalimumab is a biologic drug used to treat immune diseases such as rheumatoid arthritis (RA), psoriasis, and Crohn's disease. This project will focus on adalimumab as a treatment for RA. Adalimumab is a part of a family of drugs that inhibit tumor necrosis factor α (TNFα), a cytokine that is broadly involved in immune responses and other important cellular processes such as proliferation and cell death. Understanding the mechanism of how TNFα is regulated and how it regulates immune function is critical to the treatment of autoimmune conditions that involve this cytokine.

The following research investigates genetic sources of anti-TNF therapy such as adalimumab through several model discovery methods. High dimensional data is reduced for better interpretability using principal component analysis followed by K-means clustering to identify which genes are correlated in their clinical responses to adalimumab administration in RA patients. Only small differences were found in clustering results between the set of all patients and the set of only patients who responded positively to anti-TNF therapy, suggesting a weak association between genetic triggers and therapeutic responses. Bayesian network analysis using reduced dimensions explored causal relationships between principal components and therapeutic responses, finding that a lack of response can cause expression levels of sets of genes to change. The patterns discussed here can be further explained with gene function mapping to increase the interpretability of reduced dimensions and identify biological systems in which they are involved.

Introduction and Background

The existence of TNF's role in autoimmune disease is well established. TNF cytokines have been shown to have a role in inflammatory disease by binding to their receptors (TNFRs) and activating processes like cell death, even when they are not indicated (1, 2). Figure 1 gives a high-level view of TNFα's mechanism of action in the immune system. Soluble TNFα (sTNFα) binds to its receptor (TNFR1 or TNFR2) which in turn recruits complexes that trigger pathways leading to a variety of cell outcomes such as growth, death, and inflammation (3). In RA, overproduction of TNFα activates synovial fibroblasts in joint connective tissue that contribute to joint erosion (3, 4, 5). Adalimumab is a monoclonal antibody that inhibits TNFα by specifically binding to the cytokine and blocking cell-surface interaction with its receptors, as well as triggering apoptosis of cells that express TNFα (3, 6).

Immunogenicity occurs when protein products such as adalimumab induce immune responses against themselves by triggering the production of anti-drug antibodies (ADAs) (7). The development of immunogenic anti-adalimumab antibodies (AAAs) leaves TNF-mediated immune processes unregulated (8). AAAs neutralize the drug by binding to TNFα, preventing adalimumab binding, and allowing TNFα-TNFR interaction. While immunogenicity is known to result as a side effect of anti-TNF drugs, its genetic origin seems to be less studied. The scope of this project captures the search for co-expressed genes and genetic relationships that lead to immunogenicity from adalimumab.

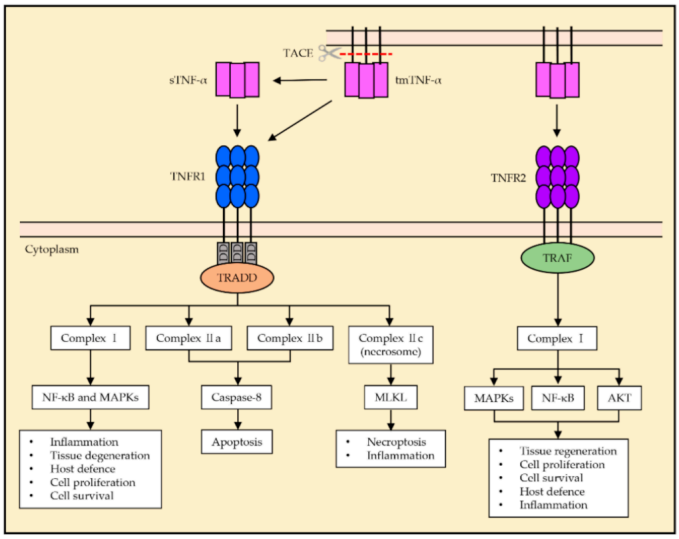

Figure 1: General TNF-α signaling pathway and possible cellular outcomes. Source: (3).

The following project is based upon work by Aterido et al. that studied transcriptomic and genomic associations with responses to anti-TNF therapy (9). They provide normalized gene expression levels from microarrays of synovial tissue for 21,184 gene probes in 11 RA patients, 3 of whom did not respond favorably to anti-TNF treatment (GEO accession GSE47726). The first of two main parts of their analysis involved finding gene co-expression modules (GCMs) of genes with correlated expression levels (Figure 2). The GCMs were then evaluated on whether or not they were associated with anti-TNF response.

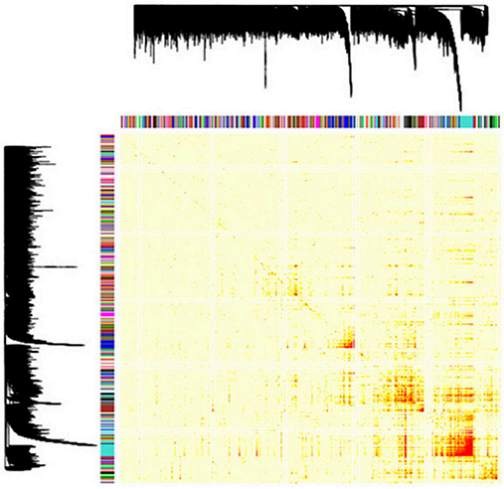

Figure 2: Gene expression modules identified by Aterido et al. Source: (9).

The following project uses machine learning methods to reduce the dimensionality of Aterido et al.'s collected data and use the transformed data to infer anti-TNF therapy responses from new data. The large feature set of the original data is reduced using Principal Component Analysis (PCA) to include only those features that contribute most to determining therapeutic outcomes. K-means clustering is then used to group the reduced features. Finally, Bayesian Network analysis is performed on the reduced features to infer causality of anti-TNF response as well as possible pathway structures involved in determining these responses. Here, these model discovery methods are employed to infer genetic causality of response to anti-TNF therapy.

Methods

The data was obtained from (9) via the Gene Expression Omnibus (GEO) database and the GEOparse python package that allows access to GEO "soft" files (10). The imported data did not require extensive preprocessing or cleaning since it gives normalized gene expression

levels for over 20,000 genes. The dataset manipulated in the following section contains expression levels of 21,189 genes in 11 RA patients.

PCA was performed using scikit-learn's PCA package to first to reduce the dimensionality of the large feature set (11). Without a concrete idea of how many dimensions would be suitable for a transformed dataset, the PCA algorithm was run to create as many components as would capture 90% of the total variance in the original dataset. A full single value decomposition (SVD) solver was used to determine directions of maximum variance by means of exact SVD.

Each principal component (PC) is defined as a linear combination of the original features (12). Contribution of each feature to each PC can be measured in several ways, and here large contributions are defined by linear combination coefficients (loadings) of large magnitude. Genes were evaluated by their significant contributions to the first PC, which captures the most variance in the data compared to other components.

K-means clustering was performed on the transformed dataset to group genes based on their "distance" from each other in 7-component space. The scikit-learn package was again utilized for its kmeans function (13). The number of clusters, hyperparameter k, was tuned with trial and error until a standard value of k = 10 was selected. Cluster labels of the transformed features were obtained and visualized in low dimensional spaces.

Bayesian network inference was performed using the python package pgmpy (14). Each patient's therapeutic response was recorded, and lack of response was included as a random variable in the Bayesian algorithm (True values for patients who did not respond to anti-TNF therapy and False values for patients who responded well to anti-TNF therapy). "Non_response" was included as the random variable in the network as opposed to "response" so that potential causes of immunogenicity may be studied. The network was found via an exhaustive search of every possible network connecting the 5 random variables (first 4 PCs plus the non_response variable). The exhaustive search method can only handle up to 6 random variables, which is why only the first 4 PCs were chosen as causal entities for the network. The first 4 PCs capture roughly 75% of the variance in the original dataset. Several scoring methods, including Bayesian Inference Criterion (BIC) and K2 score, were explored in network generation. The novel workflow of developing a Bayesian network using PCs was developed to reduce the computational expense that would arise from Bayesian inference using all the original features. The PCs are intended to have more interpretable biological function than individual genes so that broader pathways or specific genetic functions can be identified to cause immunogenic responses rather than individual genes that may have many different possible functions.

These model discovery steps were repeated for one dataset containing data from only the 8 patients who responded well to anti-TNF therapy. The analysis was also repeated for only the 3 patients who did not respond to anti-TNF therapy, but results were difficult to interpret due to small sample size. In both cases, only the PCs were used as random variables in Bayesian network inference since the response in each of the 2 groups was not varying across samples.

The microarray data in the original dataset is labelled by Illumina probe ID (prefix ILMN). In order to map these probe IDs to gene name and eventually to gene function, several databases within R Bioconductor packages were used for translation (14). Gene expression profiles were obtained using Illumina Beadchips WG6 version 2, and the corresponding database *illuminaHumanv3BeadID.db* was used for mapping.

Results

The methods described above were carried out according to the workflow laid out in Figure 3. Plots presented in the following section will be color coded according to the figure: plots describing all patients will be blue, plots describing only the responders will be green, and plots describing only the non responders will be pink.

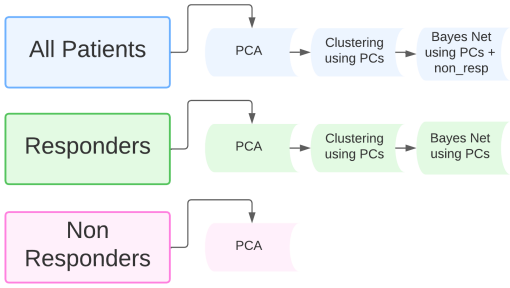

Figure 3: Model discovery workflow. There are 11 samples in the all patients group, 8 patients in the responders group, and 3 patients in the non responders group.

Model discovery of correlated therapeutic responses

PCA of the dataset with all patients was conducted such that at least 90% of the variance in the original dataset would be captured by the transformed dataset. The resulting reduced dataset had 7 principal components (PCs) that captured 90.65% of the total variance in the original dataset. The variance captured by each component is shown in Figure 4a. While 7 dimensions is deeply reduced compared to the original 21,189, it is still a difficult space in which to visualize the data. Figure 4b shows the data projected onto only the first 2 PCs. This plot shows some slight structure and deviation from complete randomness. Analogous plots for the 2 smaller datasets are shown in Figure 5a-d. The projection of transformed data onto 2 PCs again shows structure whose sources will be further investigated with clustering and Bayesian inference. To capture at least 90% of the total variance in the responders dataset, 5 PCs were made. To reach the 90% threshold with a transformed non responders dataset, only 2 PCs were made since the original data can only have a maximum of 3 components (number of components must be between 0 and min(number of samples, number of features)). All of the variation in the original non responders dataset can be explained using only these 2 PCs, but the sample size of this data is too small to make meaningful conclusions. Analysis of only non responders is omitted from further investigation to focus on the transformed datasets containing all patients and responders only.

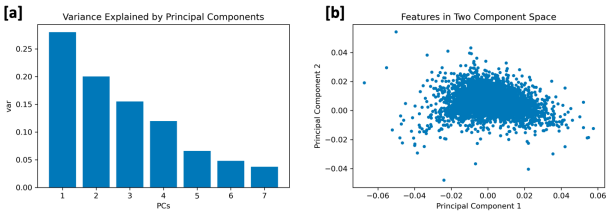

Figure 4: [a] Variance in original data captured by each PC. [b] Features projected onto first 2 PCs.

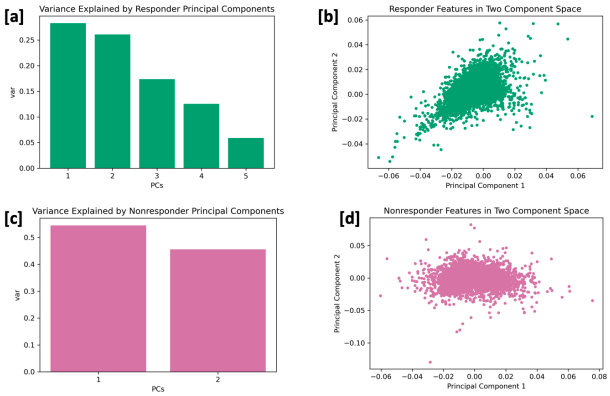

Figure 5: [a, c] Variance in original datasets captured by each PC for responders and non responders, respectively. [b, d] Features projected onto first 2 PCs..

Loadings of each gene on the first PC in each transformed dataset were examined to identify which features contribute most to PC1 in each case. The top 20 contributors to the first PC in each case are shown in Figure 6. Of each set of 20 genes, 10 are common (probe IDs: 'ILMN_8119', 'ILMN_19033', 'ILMN_4323', 'ILMN_1300', 'ILMN_27731', 'ILMN_6827', 'ILMN_26214', 'ILMN_20354', 'ILMN_1735', 'ILMN_119833').

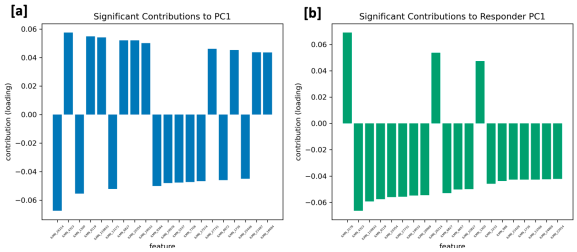

Figure 6: [a] Top 20 highest contributing genes to PC1 for all patients. [b] for responders only.

The reduced dimensions were then used in the K-means clustering algorithm to group features by their coordinates in each PC direction. Each data point was grouped into one of 10 clusters, 2D projections of which are shown in Figure 7a-b. The nature of K-means clustering groups data points based on physical distance in their 7- or 5-dimensional space, which means some commonality in clustering results between 2 datasets that largely overlap is expected. The shape of the scatterplot for responders appears to be very similar to the one for all patients if it were rotated 45 degrees counter-clockwise. The cluster labels remain largely in the same positions. Looking at cluster labels of each gene feature, there is significant overlap of which probe ID falls into each analogous cluster in the 2 datasets.

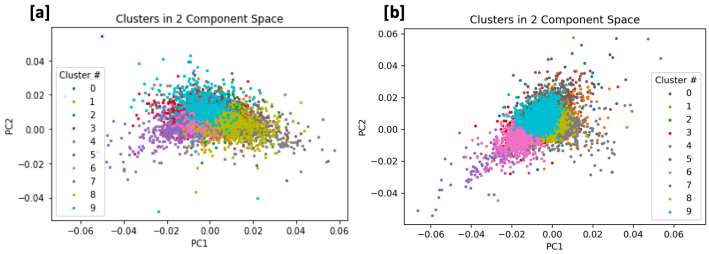

Figure 7: [a] Clusters in 2 PC space for all patients. [b] for responders only.

Genetic Topology Inference

The PCs in the full dataset were used in Bayesian network analysis to infer a genetic topology from 11 patients and their responses. The resulting network in Figure 8 shows causal relationships between the first 4 PCs and a lack of response to anti-TNF therapy. The causal relationships seems to follow the order of variance captured, meaning the PC that captures the most variance (PC1) has the most outgoing arrows, PC2 has a combination of incoming and outgoing arrows, and PC3 and PC4 have only incoming arrows from the other random variables. It was expected that the non response variable would have incoming arrows from the PCs, suggesting that expression of certain groups of genes or pathways would explain immunogenic responses to therapy. The non response variable instead only has outgoing arrows to the other PCs, suggesting that a lack of response "causes" expression of certain gene groups.

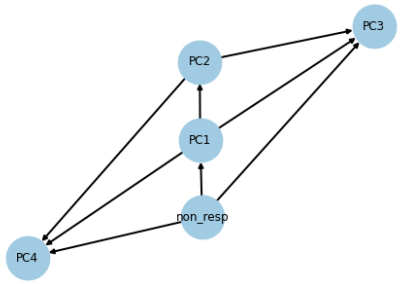

Figure 8: Bayesian network connecting the first 4 PCs of all patients and a lack of response.

A Bayesian network was also constructed from the transformed responders dataset to infer a genetic topology only among patients who responded positively to anti-TNF therapy. The resulting network in Figure 9 shows the causal power of PC1, as its outgoing arrows are the only connections in the network. PC1 accounts for 27% of the total variation in gene expression levels of responding patients and seems to be made up of a set of genes that act upstream of the gene functions represented by the other PCs in this network.

It was intended to name these PCs based on the functions of the genes that significantly contribute to them. To try to solve this task, the workflow and database developed by Barbosa-Marais et al. was followed (14). The probe IDs of significant PC contributors could not be mapped within the *illuminaHumanv2BeadID.db* database. Other related databases were also investigated to see if the probe IDs could be found in other locations. The databases are perhaps out of date since they were developed in 2009, so other up-to-date methods for mapping probe IDs to gene names should be explored.

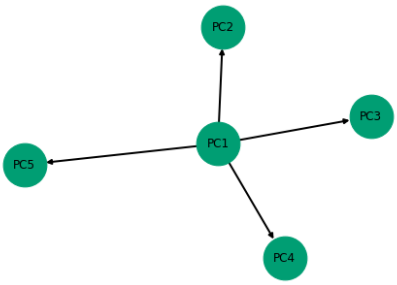

Figure 9: Bayesian network connecting first 5 PCs of responding patients.

Discussion

Conclusions

Dimensionality reduction of the full and responder datasets showed significant overlap between genes that contribute most to each PC. The commonality could indicate the role of a shared set of genes in both responding and non responding patients where response to therapy depends on inverse expression of that gene set. K-means clustering also revealed significant overlap between which probe IDs belong to which cluster in both the full and responder datasets. This similar structure in the datasets and the observation that the calculated high dimensional distances between data points did not change significantly with the removal of non responding patients suggests weak association between the expression of transformed features and a patient's therapeutic response. Those causal relationships, while potentially weak, were investigated further with Bayesian network inference. The Bayes net showed the first 4 PCs "causing" each other in order of the amount of original variance they captured, as well as a lack of response to anti-TNF therapy "causing" expression of genes contributing most to those PCs.

The next step in this investigation would be to name the PCs used throughout model discovery by mapping probe IDs to gene names and learning their functions. Doing so would give biological meaning to each PC allowing much deeper investigation into the physical sources of immunogenic responses. Clusters identified by the K-means algorithm could be compared to the GCMs identified by Aterido et al., and inconsistencies could be evaluated for novel genetic explanations for immunogenicity. Gene function information would also help identify the pathways and upstream/downstream action illustrated by the causal relationships in the Bayesian networks. Once some biological systems are identified or hypothesized to be specifically involved in immunogenicity, methods such as ODE modelling can be employed to explore the behavior of these novel systems in relation to drug response.

Project Reflection

This project largely lined up with the aims I laid out in my proposal. The model discovery process was executed smoothly with the data available to me, and I was able to learn a lot about that data by building on modeling tools from class. I was not able to learn as much biologically as I had originally hoped due to the technical challenges presented by the format of the microarray data. In future projects dealing with data obtained online, it would be helpful to thoroughly investigate mapping tools suited to a dataset before working with it so that more concrete biological explanations can be made in addition to identifying patterns that can be learned by only viewing the probe IDs.

Supplementary Information

The following files are contained in Furlow_Supplement.zip.

`GSE47726_series_matrix.csv`
Data file downloaded from GEO database to be process in `processing.py`.

`processing.py`
This file reads a series_matrix csv file downloaded from GEO GSE47726 and parses it such that the table data can be exported into a csv file containing only gene expression profiles.

`modelDisc.py`
This file performs PCA, K-means clustering, and Bayesian network inference on the set of all 11 patients.

`modelDiscRN.py`
This file performs PCA, K-means clustering, and Bayesian network inference on the 2 sets of only responders (8 patients, denoted by "R" suffix) and only non responders (3 patients, denoted by "N" suffix).

`mapping.R`
This file attempts to map Illumina probe IDs to gene names using databases developed by Barbosa-Marais et al, available through the R Bioconductor package.

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